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SURVEY

Beyond the Trends: Evolution and Future Directions in Music Recommender Systems Research

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ABSTRACT The study of Music Recommender Systems (MRS) has become crucial in digital music consumption, influencing how people discover and interact with music. This comprehensive analysis examines the complex field of MRS research from 2005 to 2023; with the growing importance of music recommendation systems in enhancing user experience, it is crucial to comprehend their development. By utilising rigorous social network analysis, statistical measures, and factor analysis, our investigation not only identifies essential themes and influential contributors but also emphasises the complex and diverse nature of MRS. The field trend significantly increased between 2017 and 2021, with periodic oscillations highlighting its dynamic nature. This analysis offers a broad perspective by examining highly cited articles, current researchers, and local sources. Factorial analysis uncovers thematic clusters, highlighting collaborative filtering, user experience, emotion identification, and reinforcement learning. A scientific mapping analysis classifies research themes in different historical periods, focusing on essential areas such as collaborative filtering, hybrid recommendation, sentiment analysis, and emotion identification. A review of thematic evolution highlights the importance of digitalisation, emotion recognition, personalisation, user experience, and collaborative filtering in determining future research directions. Although there has been a recent decrease in general interest, investigating context-aware models and hybrid techniques offers encouraging opportunities for further inquiry. This research enhances our comprehension of MRS dynamics, leading to future improvements and developments in the field. Ultimately, it improves the music discovery experience for people globally.

INDEX TERMS Music recommender system, thematic evolution, future trends, social network analysis, factor analysis.

I. INTRODUCTION

The realm of Music Recommender Systems (MRS) has witnessed a significant surge in recent years, driven by the emergence and triumph of online streaming services. These platforms now grant users access to an extensive global music library. While contemporary Music Recommender Systems effectively assist users in navigating vast music catalogues, the research landscape encounters notable challenges. Notably, developing, integrating, and evaluating recommendation strategies that delve beyond simple user-item interactions or content-based descriptors pose a substantial undertaking, resulting in a scarcity of related publications [1]. The burgeoning popularity of music streaming services has led to an exponential growth in musical content generation. This surge, however, brings about user confusion, risking a compromised quality of service in the absence of robust MRS.

It's safe to say that a recommendation service holds a crucial position in the music industry. These music recommendation services uncover similarities in music preferences between two customers who have already listened

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to particular songs. Yet, crafting a music recommendation model that introduces fresh music suggestions tailored to customers' unique preferences is undeniably challenging. Designing a model that accurately aligns with what customers love poses a real puzzle in music recommendations [2]. The primary objective of recommendations is to furnish users with meaningful suggestions based on their past interests and behaviours [3]. As the production of musical content escalates and the user base of music streaming services expands rapidly, the significance of Music Recommender Systems becomes increasingly evident. Although many existing systems rely on metadata such as genre, artist, and album, they often overlook crucial aspects like context, which holds excellent value for consumers.

The escalating quantity of tracks available on online streaming platforms underscores the crucial role of personalised MRS in identifying potential tracks and ensuring their distribution to the appropriate users. Collaborative Filtering (CF) is a frequently employed technique that dedicates item similarities for recommendation purposes. This method can be framed as a deep neural network to model user-item interactions, thereby enhancing the performance of recommendations. By implementing such a recommendation system, as exemplified by YouTube DNN, User Embedding (UE) can be derived as a function of the user's historical data and context [4]. Deep Neural Networks (DNNs) have found widespread application in collaborative-filtering-based recommender systems, showcasing notable superiority over traditional models. However, a standard limitation arises when user-item interactions are sparse in deep CF-based models, as preferences and characteristics are primarily inferred from observed historical interactions [5].

Music recommender systems have become increasingly popular because they can address the challenge of recommending relevant and novel music to users based on their musical tastes [6]. These systems have greatly benefited from algorithmic advances in the recommender systems community, such as CF, content-based, and hybrid approaches [7]. Additionally, extensive efforts have been dedicated to developing context-aware music recommender systems (CAMRSs) in recent years, aiming to improve the accuracy of recommendations by incorporating contextual information, such as location [8], [9].

Integrating user aspects into music recommendation algorithms has been explored, with proposed hybrid matching methods to recommend music for places of interest [10]. Furthermore, emotion-based music recommendation by association discovery from film music has been investigated, combining content-based and CF techniques [11]. It has been noted that most commercial music recommender systems recommend music items based on the number of plays or explicit song ratings [12].

Recent research has also focused on improving the predictive ability of top-N recommender systems by considering dimensions as virtual items [13]. Moreover, the development of deep temporal neural music recommendation models utilising music and user metadata has been proposed to address the relatively low performance of music recommendation due to the lack of a natural combination of music characteristics and users' temporal preferences [14].

In addition to traditional recommendation algorithms, new approaches have been explored, such as a personalised recommendation algorithm based on the Spark platform, aiming to address the shortcomings of conventional recommendation algorithms in dealing with large-scale music data [15]. Furthermore, the construction of a personalised recommendation system for pop music based on extensive data analysis has been investigated. This system focuses on calculating the similarity between music based on internal characteristics such as melody and beat [16].

The importance of personalised recommendation systems in the music field is evident, with various studies focusing on improving recommendation accuracy, diversity, and timeliness [17]. Additionally, emotion-oriented music recommendation algorithms fusing rating and trust have been explored to enhance the personalisation of music recommendations [18].

MRS has become an integral part of the modern music industry, aiding users in navigating the overwhelming amount of music available at their [19]. These systems are designed to help users discover new music that aligns with their preferences, creating commonality with others [20]. However, while MRSs traditionally serve well for mainstream music listeners, they often fall short in recommending music beyond the mainstream, neglecting non-popular music enthusiasts [21]. To address these challenges, researchers have proposed various innovative approaches to enhance the accuracy, diversity, and timeliness of music recommendations.

One approach involves fusing gene features in personalised music hybrid recommendation algorithms, significantly improving recommendation accuracy and precision. This ensures diverse content recommendations with higher hit rates [17]. Additionally, collaborative filtering-based music recommendation algorithms in Spark architecture have been developed to recommend music of interest to users more efficiently than traditional algorithms [22]. Furthermore, deep learning algorithms have been explored to analyse human motion and incorporate additional information for automatic music recommendations, although such studies are relatively scarce [23].

In the quest for more effective music recommendations, big data analysis, convolutional neural networks, and IoT architecture have been proposed to build intelligent music recommendation systems, aiming to provide personalised recommendations based on user tastes and emotions [24]. Moreover, the incorporation of music and user metadata in deep temporal neural music recommendation models has been explored to improve the performance of music recommendations [14]. Additionally, fuzzy inference engines have been integrated into music recommendation systems to

consider user activities and emotions as part of the recommendation parameters, addressing existing system gaps [25].

Despite these advancements, music recommendation systems still face challenges. For instance, the design of personalised music recommendation systems based on directed tags aims to provide essential music services to users and push personalised music recommendation lists, but further research is needed to enhance the effectiveness of such systems [26]. Moreover, constructing a personalised recommendation system for pop music based on big data analysis highlights the importance of considering internal song characteristics such as melody and beat to calculate music similarity for recommendations [16].

The literature review highlights the diverse array of technologies and methodologies employed in developing MRS. From CF to deep learning and fuzzy inference engines, researchers are dedicated to enhancing music recommendations' accuracy, diversity, and personalisation. The exploration extends into the future, leveraging advanced time-series models and sophisticated machine-learning algorithms to speculate on upcoming trends and shifts in the field.

The surge in MRS is not just a technological advancement but also a transformative impact on the music industry and user experience. With online streaming services dominating the music landscape, these systems play a crucial role in helping users navigate vast global music libraries. This societal impact is evident in the democratisation of music discovery, allowing listeners to explore diverse genres and discover non-mainstream tracks, fostering a more inclusive music culture. The applications of MRS extend beyond individual enjoyment to influencing music consumption trends, shaping artist visibility, and contributing to the economic dynamics of the industry.

The remainder of the paper unfolds as follows: Section II begins with a comprehensive review of the literature and explores related works. Section III introduces the methodology utilised, elucidating the intricacies of data collection and preprocessing. In Section IV, the paper navigates through applying basic statistical measures to the pre-processed data, shedding light on their consequential impact. Proceeding to Section V, the study incorporates science mapping analysis, employing techniques such as clustering to unveil patterns and insights derived from the data. Finally, Section VI encapsulates the main approaches undertaken in this research, providing a nuanced perspective on the subject matter. This section serves as the platform for concluding the paper, summarising key findings, and presenting the study's broader implications.

II. LITERATURE REVIEW

Considerable academic investigation has been devoted to music recommendation systems. This segment aims to offer a concise summary of influence.

Efforts are made to explore studies and methodological approaches within this field. Furthermore, there is a commitment to developing a sophisticated framework. That outlines essential areas, thematic dimensions, and avenues for further research, particularly emphasising CF as a pivotal and indispensable focal point in the literature review.

A. COLLABORATIVE FILTERING AND CONVOLUTIONAL NEURAL NETWORKS

Collaborative filtering exclusively operates on data associated with user-item interactions, such as explicit scoring (e.g., on a rating scale) or implicit feedback (e.g., statistics on the frequency of music piece plays or rejections). Typically, CF models predict whether a user will engage with a previously unheard music piece. This is achieved by analysing the user's past interactions and those of similar users [27].

Recommender systems, crucial tools in modern e-commerce, analyse user preferences to generate a tailored list of products. These systems aim to calculate a scoring function considering user-user and item-item similarities [28]. Two methods for similarity calculation exist: item-oriented and user-oriented [29] Bedroll demonstrated that the case-based method outperforms the user-based method based on accuracy and computational complexity [29], [30].

Despite the success of neural networks in various domains like speech recognition and natural language processing, their application in recommender systems has received relatively less attention. In a notable study [31], researchers introduced ANCF, a neural network that replaces the conventional collaborative filter's inner multiplication with a neural architecture.

However, one drawback of CF strategies is the cold start problem, particularly when suggesting items for new users or new options. The system lacks information about the preferences of new users, hindering its ability to provide informed recommendations [32].

In recent years, the rapid expansion of digital music content on the internet has led to a surge in the development of similarity and recommendation systems to manage this vast volume efficiently. As highlighted in [33]. These systems often have their foundation in music similarity mechanisms, which generate personalised playlists based on shared features among music tracks.

The utilisation of CNNs in music recommendation systems has gained prominence, as evidenced by the work presented in [33]. This approach classifies music genres based on the beats extracted from audio signals, demonstrating the efficacy of CNNs in discerning intricate features within music. Incorporating a CF recommendation algorithm further enhances the system's capabilities, merging CNN's output with user log files to provide tailored music recommendations.

Furthermore, [34] this study contributes to the literature by proposing a music genre classification system utilising CNNs for extracting high-level features from intermediate network layers. It emphasises the importance of properly managing the massive volume of online music content, essential for optimising music recommendation systems

and improving user engagement. Additionally, the authors employ cosine similarity and Euclidean distances for similarity measurement, showcasing a comprehensive approach to music recommendation.

B. MUSIC INFORMATION RETRIEVAL

Exploring Recent Advancements and Applications in Music Information Retrieval (MIR) delves into the evolving field of Music Information Retrieval, offering a comprehensive overview of its latest developments. Notably, the book focuses on emerging trends such as semantic auto-tagging, user-centric retrieval, and recommendation strategies within MIR. The initial sections provide an extensive review of established methods for feature extraction and music indexing, drawing insights from audio signals and contextual data sources like web pages and collaborative tags. These methods lay the found, user-centric retrieval, and rival tasks, including semantic music search and query by example [35].

Traditional information retrieval (IR) systems, primarily based on text information, are inadequate for multimedia retrieval due to the diverse nature of multimedia content. Specifically, MIR systems have explored innovative approaches, such as query-by-humming, where users input melodies for retrieval. However, challenges persist in achieving precision and effectiveness, requiring advancements in melody representation, matching methods, and pitch extraction. This literature review underscores the existing issues in conventional MIR systems. It introduces the concept of a dynamic melody representation to enhance MIR systems' robustness and efficiency. The proposal addresses singing errors, event detection, and pitch extraction challenges, emphasising the need for a comprehensive and error-tolerant approach for a successful MIR system [36].

C. COLD START PROBLEM

Among the challenges recommender systems (RSs) face, the cold start problem is persistent. This problem arises when RSs lack sufficient information about users or items, making providing accurate and relevant recommendations challenging. The existing literature offers various solutions to tackle this problem, categorised into two classes based on the nature of collecting missing information: explicit and implicit solutions.

Researchers actively collect missing information in the explicit solutions category to enhance recommendation accuracy. This involves strategies such as soliciting user feedback through surveys or ratings or gathering additional item data from external sources. The work presented in [32] This approach is exemplified by proposing a model that addresses the new user cold start problem. The model incorporates classification algorithms, similarity techniques, and prediction mechanisms to provide user recommendations without historical preference data. Noteworthy is the emphasis on handling the cold start problem without relying on a priori probability, complex calculations, or the need for user interviews.

Implicit solutions, on the other hand, explore alternative methods for handling the cold start problem without explicitly collecting missing information. The text discussed [37] Provides an insightful overview of the cold start problem in RSs, highlighting the challenges associated with new users and items. The article discusses techniques such as active learning, decision trees, and social recommender systems as potential implicit solutions. These approaches leverage existing data, user demographics, and implicit feedback like browsing and purchasing histories to make recommendations without requiring explicit ratings or questionnaires.

The two classes of solutions present distinct advantages and considerations. Explicit solutions, as demonstrated in [32], each actively addresses the cold start problem by integrating classification algorithms and prediction mechanisms. On the other hand, implicit solutions, as outlined in [37], leverage existing user data and behaviours to make recommendations without needing explicit information collection.

While explicit solutions may provide more accurate recommendations for new users, implicit solutions offer a less intrusive approach that aligns with the user's natural interactions within the system. Hybrid approaches, combining both explicit and implicit strategies, may provide a balanced solution to the cold start problem, as mentioned in [37].

In conclusion, the literature review emphasises the significance of the cold start problem in recommender systems and explores diverse strategies to mitigate its impact. The explicit and implicit solutions, exemplified by the works in [32] and [37] respectively, contribute valuable insights into addressing the challenges associated with new users and items in RSS.

D. HYBRID RECOMMENDATION

A novel approach in music recommender systems, termed hybrid recommendation, amalgamates various methodologies to furnish tailored music suggestions. This method amalgamates factors such as audio signal beats, user inclinations, and log files to categorise music into genres and discern similarities among them [38], [39]. By integrating content-based filtering, collaborative filtering, and knowledge graphs, this hybrid recommendation system advances media recommendations grounded on user interactions and preferences [38], [40]. Unlike conventional recommendation algorithms, this approach leverages music genes, user and item label data, and deep learning techniques to surmount their limitations [41]. Specifically, the hybrid model, RCTR-SMF, synergises social matrix factorisation and collaborative topic regression to tackle sparsity issues and enhance prediction accuracy in music artist recommendations [41].

E. CONTEXT-AWARENESS

Context awareness in music recommender systems entails leveraging contextual cues to deliver personalised recommendations aligned with users' preferences and the circumstances of their music-listening experiences. This entails extracting

contextual insights from users' listening histories, encompassing attributes like tags, genres, and metadata, and leveraging them to enrich song embeddings [42]. Moreover, sophisticated feature selection algorithms, such as those rooted in genetic algorithms, facilitate compressing high-dimensional contextual data into more concise latent spaces. This not only enhances recommendation accuracy but also fosters interpretability [43]. Introducing context-aware explanations within recommender systems further bolsters transparency, instils trust, and boosts user satisfaction [44]. Additionally, in multi-stakeholder platforms, employing contextual bandit algorithms proves instrumental in considering the preferences of all involved parties, thereby enabling the provision of recommendations that cater to diverse stakeholders over time [45].

F. AFFECTIVE COMPUTING

Affective computing plays a pivotal role in enhancing the effectiveness of music recommender systems by incorporating contextual factors such as user emotion and uncertainty into the recommendation process. This entails leveraging effective analysis technologies, including physical and physiological signals, to gauge users' emotional responses. Physiological signals, such as heart rate, electrodermal activity, and brain activity, offer reliable insights for emotion analysis [46]. Central to this approach is the concept of affective information, which is utilised to construct a music-affective space. Leveraging fusion-based recommendation methods, tailored recommendations for serendipitous music exploration are provided [47]. Furthermore, user personality and mood recognition are seamlessly integrated into content-based filtering approaches, augmenting recommendation accuracy [48]. Moreover, computer vision components discern users' emotions via facial expressions. This innovative technique enables personalised song suggestions aligned with users' prevailing states of mind [49].

G. PERSONALIZATION

Personalisation within music recommender systems is pivotal for tailoring recommendations to individual user preferences and behaviours. A range of methodologies has been proposed to achieve this. One such approach involves leveraging user-centric artificial intelligence, such as enhanced CF. CF addresses sparse data challenges by incorporating user actions beyond explicit ratings, thereby generating high-quality recommendations [50]. Another effective strategy revolves around disentangling the latent factors embedded within user-item interactions to unveil their underlying dependencies. Techniques like variational inference and latent structure learning are deployed to replicate observed interaction patterns and meet causal prerequisites [50]. Additionally, content-based filtering and unattended learning models offer insightful analysis of specific playlist features. This allows for recommendations to be provided categorised by genre, artist, and even mixed suggestions [51], [52]. The overarching objective of these personalised music recommendation systems is to elevate user satisfaction and optimise recommendation effectiveness, thus enriching the overall listening experience.

H. MUSIC CLASSIFICATION

Music classification is a cornerstone in music recommender systems, yet traditional machine learning techniques struggle to grasp the intricacies of music data fully [53]. While deep learning architectures, such as Convolutional Neural Networks, excel at capturing spatial hierarchies, they often falter at capturing the temporal dynamics inherent in music [54]. A novel hybrid model has emerged, melding the strengths of Residual Neural Networks (ResNet) and Gated Recurrent Units (GRU) to tackle these challenges head-on [55]. Moreover, machine learning algorithms like XGBoost have been harnessed to automatically detect music genres, thus facilitating the development of robust recommendation systems [56]. Conversely, deep learning models like Feedforward Neural Networks (FNNs) have showcased superior predictive capabilities in music classification tasks [57]. Additionally, leveraging feature extraction techniques alongside neural networks has proven fruitful in crafting successful music retrieval systems. These multifaceted approaches collectively aim to bolster the accuracy and comprehensiveness of music genre classification, ultimately enhancing the user experience within music recommender systems.

I. EMOTION RECOGNITION

Emotion recognition is a pivotal element within music recommender systems, providing personalised music suggestions that resonate with users' emotional states. Facial expressions serve as a standard indicator of mood, and advancements in technology facilitate the extraction of discernible facial traits as inputs for emotion recognition. Transfer learning is a powerful technique in enhancing emotion recognition and music recommendation systems, leveraging prior knowledge to diminish the reliance on labelled data. Several models, including CNNs, ResNet50V2, VGG16, and EfficientNet50 B0, have been employed for emotion detection. These models undergo training on datasets such as FER2013 and AffectNet. In tandem, music recommendation systems utilise clustering algorithms like k-means to group tracks based on predicted emotions, thus furnishing song recommendations aligned with users' emotional states. The integration of emotion recognition into music recommender systems not only enhances the accuracy of emotion detection but also elevates the overall user experience, fostering deeper engagement and satisfaction [50], [58], [59], [60], [61].

J. USER EXPERIENCE

Traditional music recommender systems have historically prioritised prediction accuracy over user experience (UX) considerations when generating recommendations. However, recent research has shed light on the pivotal role of UX in the

effectiveness of recommendation systems [62], [63]. Factors such as injecting novelty into recommendations, ensuring regular updates, and catering to users' interest in rating significantly foster a positive UX [64]. Furthermore, the dynamics of interaction between music listeners and platforms, coupled with elements like satisfaction levels, user activity, and feedback, exert considerable influence on UX and shape listening preference [65]. Notably, user-centred evaluations of tour-generation algorithms offer valuable insights into optimal music presentation methods and avenues for enhancing recommender systems [66]. Moreover, investigations into the evolution of users' musical preferences and their exploration behaviour within genres underscore the importance of striking a delicate balance between personalisation and exploration to cultivate a helpful and gratifying user experience.

K. USER AND ITEM-ORIENTED

Two primary approaches are employed in determining similarity within music recommender systems: item-oriented and user-oriented methods. Item-oriented methods focus on gauging the relevance between items, such as songs, based on their intrinsic characteristics or features [67]. Conversely, user-oriented methods estimate relevance between users by analysing their behaviours and preferences, offering recommendations aligned with similar users' musical tastes [68]. These approaches are essential for delivering precise and personalised music recommendations, enriching the music-listening experience.

Item-oriented similarity methods encompass techniques like content-based filtering, which directly compares the similarity value of a song with the user's favourite, predicting songs with high similarity values [69]. Content-based recommendation algorithms further suggest items with significant similarity based on historical selection behaviours that users have appreciated [70]. Additionally, predefined and learned similarity methods are employed to compute the similarity between items, subsequently identifying items of potential interest based on the frequency of their occurrences [68].

On the other hand, user-oriented similarity methods utilise CF to deduce similarities between items for recommendation [4]. CF methods depend on explicit rating values to identify similar users or items. They commonly predict a user's interest through their listening habits and similarities with other users' profiles [71]. Furthermore, personalised relationships-based knowledge graphs are applied to explore user-item relatedness by extracting associated entities from user-oriented and item-oriented entity views, enhancing item semantic information [72].

In summary, item-oriented and user-oriented similarity methods play a pivotal role in the effectiveness of music recommender systems.

III. METHODOLOGY AND DATA

We conducted a bibliometric analysis using the Bibliometrix R-package [73], [74]. A comprehensive scientific mapping analysis will be performed in five stages. First, we collected and downloaded raw data related to Music recommender systems from the Web of Science, following the guidance of prior studies [75], [76]. After data collection, we proceeded with data cleaning and initial statistical analysis to understand fundamental document properties, such as publication trends over the years. In the third stage, we preprocessed keywords from the documents, grouping similar and equivalent terms and constructing a co-occurrence matrix for unit analysis. Following this, we applied data and network reduction filters to streamline the analysis and employed a clustering algorithm inspired by prior research [74], to generate maps, clusters, and subnetworks. Finally, we created various visual diagrams, including evolution maps, strategic diagrams, thematic networks, and clustering diagrams, using the Bibliometrix R-package, which provided valuable insights into the trends and core topics in music recommender systems research projects [73], [74], [75], [76].

While the bibliometric analysis conducted using the Bibliometrix R-package provides valuable insights into trends and core topics in music recommender systems research, it is essential to acknowledge several limitations. The study's reliance on a single database for data collection may introduce coverage bias, potentially omitting relevant publications beyond the selected database's scope. Additionally, the accuracy and consistency of keywords used for data collection may introduce biases, and the quality of the analysis depends on the reliability of the source data. A publication lag exists, as the time between research conduct and publication may lead to delayed representation of the most recent developments in the field. Language bias is inherent, given the predominant focus on English-language publications. The clustering algorithm has limitations in capturing interdisciplinary or niche topics, which could result in underrepresentation. The static nature of the generated maps may not fully capture the dynamic evolution of research trends over time. Moreover, it is crucial to note that cleaning data is time-consuming. Recognising these limitations is essential when interpreting the findings, providing a comprehensive understanding of the potential constraints associated with the methodology.

Furthermore, the bibliometric analysis is subject to the limitations inherent in the chosen metrics and indicators. While citation count and h-index offer quantitative measures of research impact, they may not capture the full spectrum of a publication's influence or significance. Variability in citation practices across disciplines and the potential for self-citations can also impact the accuracy of these metrics.

The study's exclusive focus on peer-reviewed journal articles and conference papers may also exclude valuable insights from non-traditional sources such as blogs, preprints, and industry reports. The rapidly evolving nature of technology-related research, particularly in music recommender systems, may mean that significant developments and innovations are disseminated through alternative channels not captured in the analysed dataset. Recognising the limitations

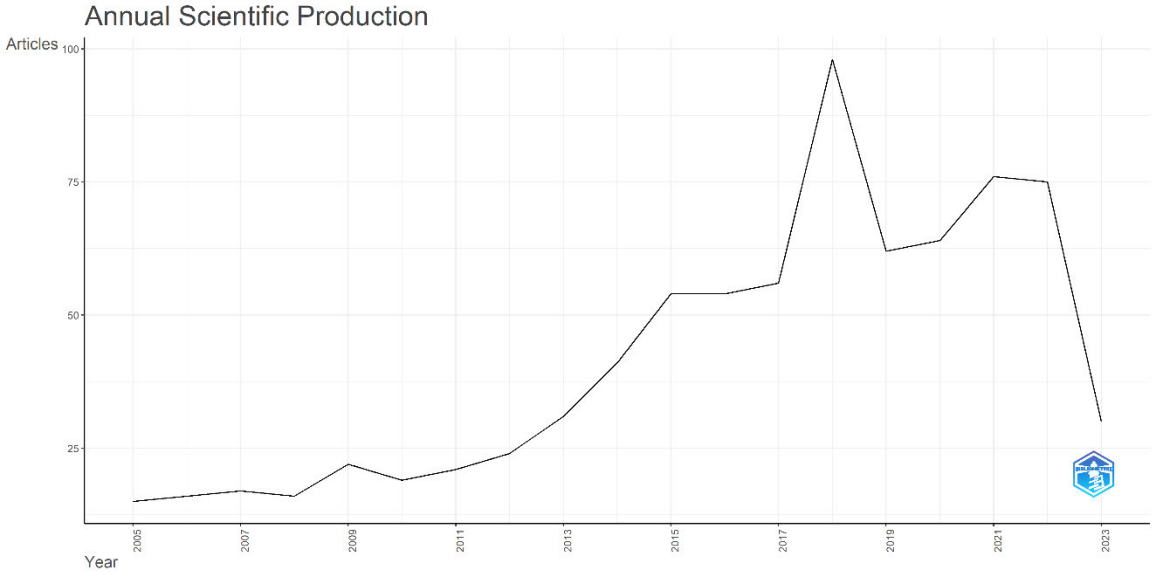


FIGURE 1. Annual number of publications related to music recommender system.

associated with these methodological choices is crucial for understanding the scope and potential biases in the findings, emphasising the need for complementary approaches to ensure a more comprehensive and nuanced perspective on the landscape of music recommender systems research.

IV. STATISTICAL MEASURE

Figure 1 indicates the yearly count of publications about the Music recommender system. From 2005 to 2023, we observe an average annual growth rate of 8.42 percentage points in the production of pertinent research documents. Notably, this surge is most pronounced from 2017 to 2018 and 2019 until 2021. On the other hand, there is a noticeable decrease in publications from 2018 to 2019. Detailed information regarding the most referenced papers within these time frames can be found in Table 1.

In the realm of publications, there were 75 single-authored documents authored by 62 individuals worldwide. On average, three authors collaborated on other works.

The entries listed in Table 1 encompass a broad spectrum of subjects and viewpoints within music recommendation systems. This diversity reflects the multifaceted complexity of these systems, exploring their potential and contradictions, effects on user choices, examination of algorithmic biases, and the need for thorough reviews and research agendas. The broad range of topics suggests that research on music recommendation systems has not followed a single path but has explored multiple dimensions and challenges associated with this ever-changing technology.

Tables 2 and 3 present a compilation of ten distinguished authors and ten highly referenced journals from 2015 to 2023. These tables highlight influential individuals and esteemed academic institutions that have played a crucial role in shaping the field of music recommendation systems.

A. FACTORIAL ANALYSIS

In this section, we will perform factor analysis to unveil latent patterns and relationships within a collection of observed variables. Our objective is to expose thematic clusters and dimensions and offer insights by interpreting tables and figures. According to the data in Table 4 and what is shown in Figure 3, we analyse the clusters:

1. Cluster 1:

Keywords: collaborative filtering, music information retrieval, convolutional neural networks, context awareness, personalisation, cluster-based, content-based filtering, experiment, matrix factorisation, music therapy, user modelling, emotion-aware music recommender, visualisation, background music, sentiment analysis, implicit feedback, music classification, music emotion recognition, recommendation algorithm.

Characteristics: A diverse set of keywords related to collaborative filtering, neural networks, content-based filtering, emotion, and various recommendation algorithms.

2. Cluster 2:

Keywords: user experience, affective computing, emotion recognition.

Characteristics: Focus on user experience, affective computing, and emotion recognition.

3. Cluster 3:

Keywords: knowledge graph, automatic playlist continuation, cold start problem, content-based recommendation, diversity, embedding, cold start, graph, fairness, novelty, reinforcement learning.

TABLE 1. Table 1 most cited publications in music recommender system.

Paper	Total Citations	TC per Year
WANG H, 2019, WEB CONFERENCE 2019: PROCEEDINGS OF THE WORLD WIDE WEB CONFERENCE (WWW 2019)	315	63.00
WANG X, 2014, PROCEEDINGS OF THE 2014 ACM CONFERENCE ON MULTIMEDIA (MM'14)	194	19.40
SCHEDULE M, 2018, INT J MULTIMEDIA INF REAR	115	19.17
CHENG Z, 2016, ACM TRANS INF SYST	110	13.75
NILASHI M, 2014, EXPERT SYST APPL	104	10.40
KAMINSKAS M, 2012, COMPUT SCI REV	103	8.58
SU JH, 2010, IEEE INTELL SYST	103	7.36
BONNIN G, 2015, ACM COMPUT SURV	99	11.00
PASSANT A, 2010, SEMANTIC WEB-ISWC 2010, PT II	92	6.57
MIERSWA I, 2005, MACH LEARN	90	4.74

TABLE 2. Top active researchers of music recommender system.

Paper	Total Citations
WANG H	315
WANG X,	194
SCHEDL M	115
CHENG Z	110
NILASHI M	104
KAMINSKAS M	103
SU JH	103
BONNIN G	99
PASSANT A	92
MIERSWA I	90

TABLE 3. Top most local cited sources of music recommender system.

Sources	Articles
MULTIMEDIA TOOLS AND APPLICATIONS	23
EXPERT SYSTEMS WITH APPLICATIONS	15
RECSYS CHALLENGE'18: PROCEEDINGS OF THE ACM RECOMMENDER SYSTEMS CHALLENGE 2018	10
INTERNATIONAL JOURNAL OF MULTIMEDIA INFORMATION RETRIEVAL	9
IEEE ACCESS	8
15TH ACM CONFERENCE ON RECOMMENDER SYSTEMS (RECSYS 2021)	7
APPLIED SCIENCES-BASEL	7
COMPLEXITY	7
ELECTRONICS	7
IEEE TRANSACTIONS ON MULTIMEDIA	7

Characteristics: Involves knowledge graphs, playlist continuation, cold start issues, diversity, and reinforcement learning.

Dimension 1: Dimension 1 encompasses various topics related to collaborative filtering, neural networks, emotion, and recommendation algorithms. It suggests that the study or literature under analysis covers diverse topics related to music recommendation systems. It indicates a focus on technical aspects like neural networks, collaborative filtering, and emotional and personalised aspects of music recommendation.

Keywords with Positive Loadings: Music information retrieval, convolutional neural networks, context awareness, personalisation, cluster-based, content-based filtering, experiment, matrix factorisation, music therapy, user modelling, emotion awareness, music recommender, visualisation, background music, sentiment analysis, implicit

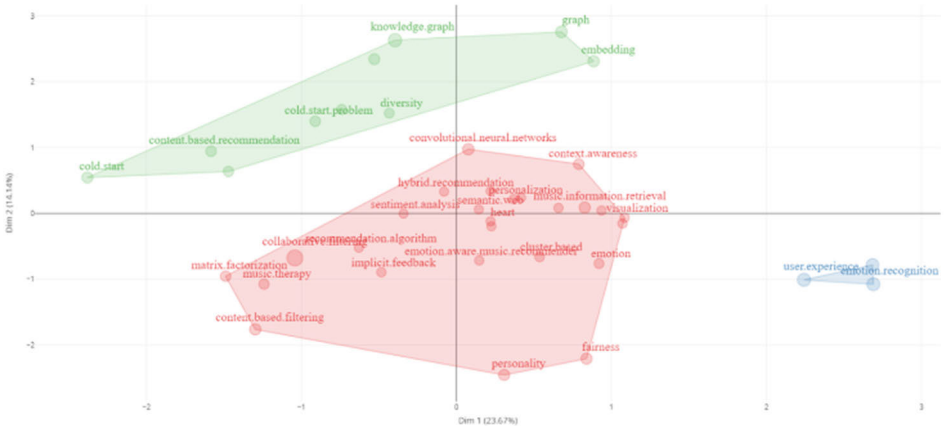


FIGURE 2. Word map of music recommendation systems research keywords.

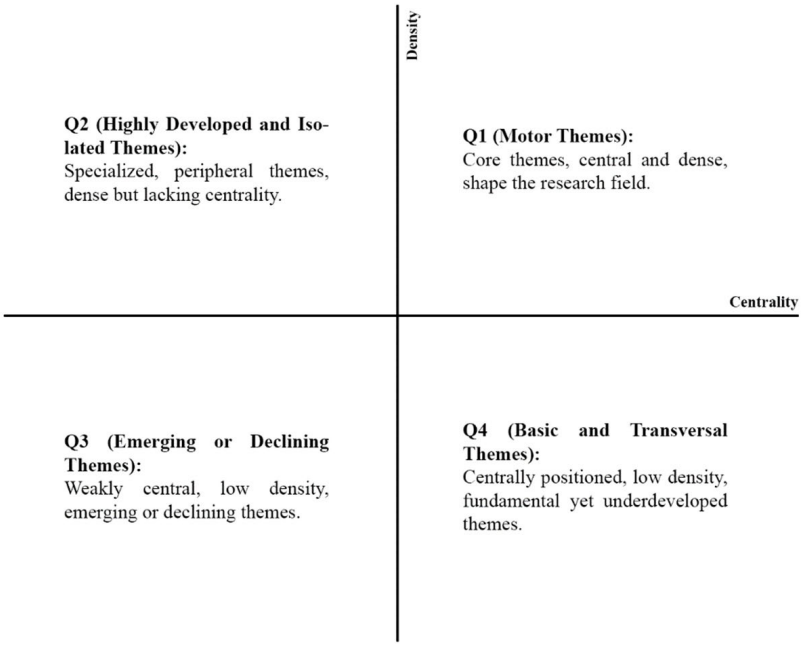


FIGURE 3. Two-dimensional strategic diagram.

feedback, music classification, music emotion recognition, the recommendation algorithm.

Emphasis: This dimension encompasses a diverse set of topics with positive loadings, including collaborative filtering, neural networks, emotion, and various recommendation algorithms.

Dimension 2: A focus on user experience, affective computing, and emotion recognition characterises dimension 2. It suggests a specific emphasis on understanding and improving user experience in the context of music recommendation systems. It indicates a focus on technologies and methodologies for recognising and incorporating user emotions into the recommendation process.

Keywords with Positive Loadings:

User experience, affective computing, emotion recognition

Emphasis: Dimension 2 focuses on user experience, affective computing, and emotion recognition.

Keyword Analysis:

- Affective Computing (Loading: 2.68):
  - *Significance:* Has a favourable loading in Dimension 2, indicating its relevance to user experience and emotion recognition.
- Reinforcement Learning (Loading: 2.34):
  - *Significance:* There is strong positive loading in Dimension 1, highlighting the emphasis on reinforcement learning in the provided keywords.
- Cold Start (Loading: -2.38):

TABLE 4. Factorial analysis of music recommendation systems research keywords.

Word	Dim.1	Dim.2	cluster	word	Dim1	Dim2	cluster
Collaborative filtering	-1.04	-0.68	1	Cold start	-2.38	0.55	3
Music information retrieval	0.83	0.09	1	Embedding	0.89	2.31	3
Convolutional neural networks	0.08	0.97	1	Fairness	0.84	-2.21	1
Context awareness	0.79	0.75	1	Heart	0.22	-0.12	1
Personalisation	0.37	0.22	1	Semantic web	0.14	0.06	1
Cluster-based	0.54	-0.66	1	Sentiment analysis	-0.34	0	1
User experience	2.24	-1.01	2	Background music	1.07	-0.15	1
Hybrid recommendation	-0.08	0.33	1	Diversity	-0.43	1.52	3
Knowledge graph	-0.4	2.63	3	Emotion detection	0.22	0.34	1
Automatic playlist continuation	-1.47	0.64	3	Graph	0.68	2.76	3
Cold start problem	-0.91	1.4	3	Implicit feedback	-0.48	-0.89	1
Emotion	0.92	-0.76	1	Music classification	0.94	0.04	1
Affective computing	2.68	-0.79	2	Music emotion recognition	0.42	0.24	1
Content-based filtering	-1.3	-1.76	1	Novelty	-0.74	1.58	3
Content-based recommendation	-1.58	0.94	3	Personality	0.31	-2.45	1
Emotion recognition	2.69	-1.08	2	Recommendation algorithm	-0.63	-0.52	1
Experimentation	0.23	-0.2	1	Reinforcement learning	-0.53	2.34	3
Matrix factorisation	-1.49	-0.96	1	Emotion-aware music recommender	0.15	-0.72	1
Music therapy	-1.24	-1.08	1	Visualisation	1.09	-0.06	1
User modeling	0.66	0.08	1	Cold start	-2.38	0.55	3
Emotion-aware music recommender	0.15	-0.72	1	Embedding	0.89	2.31	3
Visualisation	1.09	-0.06	1				

  - *Significance:* Strong negative loading in Dimension 1, suggesting a focus on addressing cold start problems in the literature.
- Fairness (Loading: -2.21):
  - *Significance:* Negative loading in Dimension 1, indicating a potential focus on fairness in recommendation systems.

V. SCIENCE MAPPING ANALYSIS

A. CONTENT ANALYSIS

Following the earlier stages, the research themes are classified within the strategic diagram of the studied period, considering their centrality, density, Callon centrality, and Callon density rankings [77].

Figure 3 depicts the overall structure of a two-dimensional strategic diagram. Centrality evaluates the extent of a network's interaction with others, while density examines and quantifies internal strength.

Aligned with this figure, the diagram can be interpreted through four distinct areas [75].

The subsequent section will delve into the four primary strategic diagrams and associated keywords for each of the defined time frames to visually represent the research themes within the field of music recommender systems for each specific period.

1) ANALYSIS OF THE STRATEGIC DIAGRAM OF THE FIRST SUB PERIOD (2005-2014)

a: MOTOR THEMES: COLLABORATIVE FILTERING

In the MRS landscape, CF emerges as a highly influential theme. This technique leverages the known preferences of a group of users to predict the undiscovered preferences of others. Utilising a database of user-item interactions, CF forecasts additional topics or products that a new user may find appealing [50].

b: HIGHLY DEVELOPED AND ISOLATED THEMES: CONVOLUTIONAL NEURAL NETWORKS, HYBRID RECOMMENDATION

Within the developmental trajectory of MRS, CNNs and Hybrid Recommendation systems stand out as highly developed and isolated themes. CNNs have showcased remarkable progress in speech recognition and large-scale image classification, surpassing traditional models. Their success can be attributed to critical elements like rectified linear units (ReLUs), parallelisation for accelerated training through

TABLE 5. Performance measures for the themes of the sub period 2005-2014.

Cluster	Callon Centrality	Callon Density	Rank Centrality	Rank Density	Cluster Frequency
collaborative filtering	0.0326	25.2564	8	9	35
music information retrieval	0.0302	5.5556	7	1	18
context-awareness	0.0642	14.2857	9	2	7
content-based filtering	0	20	3.5	4.5	5
experimentation	0	20	3.5	4.5	5
personalisation	0	20	3.5	4.5	5
visualisation	0	20	3.5	4.5	5
convolutional neural networks	0	25	3.5	7.5	4
hybrid recommendation	0	25	3.5	7.5	4

GPU acceleration, and access to substantial training data, such as the Million Song Dataset (MSD) [52], [53], [54], [55].

c: EMERGING OR DECLINING THEMES: VISUALIZATION, CONTENT-BASED FILTERING, EXPERIMENTATION, PERSONALIZATION

The strategic map for Music Recommender Systems research between 2005 and 2014 identifies emerging or declining themes. Content-based filtering, a process of aligning user profile attributes with content object attributes for recommending new items, is a noteworthy emerging theme. On the other hand, visualisation, content-based filtering, experimentation, and personalisation are themes that either show signs of emergence or decline within this period.

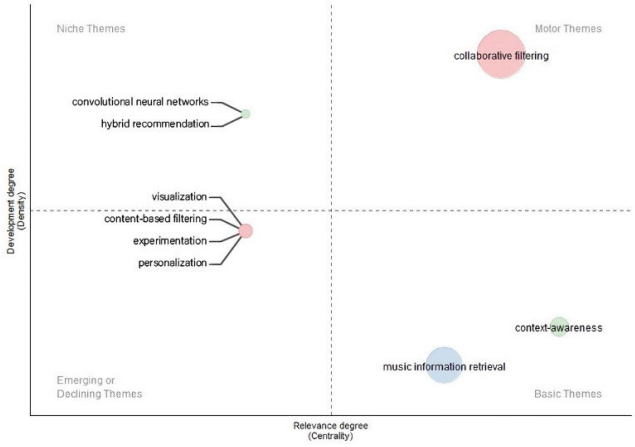


FIGURE 4. Strategic diagram for the first sub period of 2005-2014.

d: BASIC AND TRANSVERSAL THEMES: CONTEXT-AWARENESS, MUSIC INFORMATION RETRIEVAL

Basic and transversal themes in MRS encompass context awareness and music information retrieval. Context awareness plays a pivotal role, with studies analysing the contextual elements influencing recommendation systems. Music information retrieval, a fundamental theme, involves retrieving, organising, and making sense of music-related data to enhance the overall effectiveness of recommender systems.

This strategic map provides a comprehensive view of the evolution and trends in Music Recommender Systems research, highlighting innovative themes that have shaped the landscape over the specified time frame.

The following Figure (4) and Table (5) show the strategic diagram for the 222 researches which are published between the years 2005 to 2014

In MRS's developmental trajectory, the preeminent theme in its early stages is collaborative filtering. This technique draws on the known preferences of a group of users to predict the undiscovered preferences of other users, utilising a database of user-item interactions to forecast additional topics or products that a new user may find appealing [78].

A noteworthy emerging theme is content-based filtering, a process involving the alignment of attributes from a user profile, where preferences and interests are stored, with those of a content object (item). This alignment facilitates the recommendation of new, exciting items to the user [79]. Within this period's strategic map, the integration of Convolutional Neural Networks stands out as a compelling theme. CNNs have demonstrated significant progress in speech recognition and large-scale image classification, outperforming traditional models [80], [81]. Critical elements, including the use of rectified linear units (ReLUs) for faster convergence, parallelisation for accelerated training through GPU acceleration, and the availability of substantial training data, such as the Million Song Dataset (MSD), facilitate the training of large models, contribute to this approach's success [82], [83].

This strategic map provides a comprehensive view of the evolution and trends in Music Recommender Systems research, shedding light on innovative themes that have shaped the landscape.

2) ANALYSIS OF THE STRATEGIC DIAGRAM OF THE SECOND SUB PERIOD(2015-2018)

a: HIGHLY DEVELOPED AND ISOLATED THEMES: SENTIMENT ANALYSIS, HYBRID RECOMMENDATION, AUTOMATIC PLAYLIST CONTINUATION

Certain themes within MRS have attained a highly developed and isolated status. Sentiment Analysis, a notable theme, has evolved beyond textual contexts, extending its reach to include acoustic music information. This innovative approach focuses on deciphering the emotional content of music lyrics, providing an alternative to traditional CF methods. The integration of Sentiment Analysis introduces lightweight, language-independent models, contributing to the diversified landscape of MRS research [62], [63].

Hybrid Recommendation systems also stand out as highly developed and isolated themes, persisting as a niche approach that combines multiple recommendation strategies. These hybrid models strategically leverage the complementary advantages of various recommendation techniques, offering a more nuanced and practical approach to music recommendation [61].

Additionally, Automatic Playlist Continuation, while initially a highly developed theme, exhibits signs of evolution within the current MRS landscape.

b: EMERGING OR DECLINING THEMES: AUTOMATIC PLAYLIST CONTINUATION

The strategic map of MRS research highlights themes' emergence and potential evolution over time. In this context, Automatic Playlist Continuation is a theme that has changed. As the field continues to evolve, the status of automatic playlist continuation within MRS is subject to shifts, indicating a dynamic landscape with evolving trends.

c: BASIC AND TRANSVERSAL THEMES: COLLABORATIVE FILTERING, MUSIC INFORMATION RETRIEVAL

CF has transitioned from a pivotal to a fundamental theme in the current era of MRS research. The strategic map emphasises the integration of CF with MIR, a field dedicated to extracting meaningful features from music across various dimensions, including audio signals, symbolic representations, and external sources. This integration explores semantic data, context-aware systems, motion data, and geographical information, enriching recommendation strategies with a deeper understanding of user preferences [56], [57], [58], [59], [60].

These basic and transversal themes underscore the foundational role of CF and the dynamic evolution of MRS research. The integration of CF with MIR signifies a comprehensive approach beyond traditional recommendation techniques, exploring diverse data sources to enhance the precision and personalisation of music recommendations.

As evident in the strategic map, CF has transitioned from a pivotal to a fundamental theme within MRS. In the current era, there is a notable emphasis on integrating CF with MIR, a field dedicated to extracting meaningful features from music across audio signals, symbolic representations, and external sources. This integration explores semantic data, context-aware systems, motion data, and geographical information, enhancing the depth of recommendation strategies [84], [85], [86], [87], [88].

In parallel, Hybrid Recommenders persist as a niche theme, blending multiple recommendation strategies to leverage their complementary advantages [89].

A noteworthy evolution in this period is the emergence of Sentiment Analysis, or 'opinion mining,' extending beyond textual contexts to include music acoustic information. Utilising sentiment analysis to decipher the emotional content of music lyrics presents an alternative approach to CF, focusing on developing lightweight, language-independent models [90], [91].

This consolidated view underscores the dynamic evolution of CF and the diversification of themes in MRS research, showcasing its foundational role and the emergence of innovative approaches.

3) ANALYSIS OF THE STRATEGIC DIAGRAM OF THE THIRD SUB PERIOD (2019-2020)

a: MOTOR THEMES: COLLABORATIVE FILTERING, PERSONAL CHARACTERISTICS

In the context of Music Recommender Systems, the motor themes quadrant emphasises the integration of CF and the exploration of personal characteristics. CF remains a pivotal approach in the early stages of MRS, leveraging user interactions to make accurate recommendations. Simultaneously, the spotlight is on incorporating personal characteristics into the recommendation process. This involves a comprehensive analysis of user attributes, demographics, and psychographics to tailor recommendations based on individual traits, adding sophistication to the user-centric recommendation experience [67].

b: HIGHLY DEVELOPED AND ISOLATED THEMES: LYRICS, COLD-START PROBLEM, CLASSICAL MUSIC, MUSIC PREFERENCES, CONTEXT-AWARENESS

During the initial phases of MRS development, the focus revolves around highly developed and isolated themes. Addressing the cold-start problem and incorporating context awareness emerges as a dominant theme. The cold-start problem challenges accurate recommendations for new users or items with limited interaction history, prompting researchers to leverage auxiliary data sources and contextual information for more precise recommendations. Additionally, specific themes such as lyrics, classical music, and music preferences

gain prominence, indicating a diversified landscape with focused areas of exploration.

c: EMERGING OR DECLINING THEMES: PERSONALIZATION, MUSIC CLASSIFICATION

Personalisation takes centre stage among emerging or declining themes. Researchers are actively refining and advancing personalisation techniques, exploring novel algorithms to provide users with highly tailored recommendations. This phase witnesses a growing interest in understanding user context, behaviour, and preferences for more nuanced and accurate recommendation systems. On the contrary, music classification is a theme that might be experiencing shifts, highlighting the dynamic nature of trends within MRS research.

d: BASIC AND TRANSVERSAL THEMES: MUSIC INFORMATION RETRIEVAL, CONVOLUTIONAL NEURAL NETWORKS, COLLABORATIVE FILTERING

Basic and transversal themes play a fundamental role in shaping the MRS landscape. Music Information Retrieval stands out as a cornerstone theme involving extracting meaningful features from diverse music data sources. Convolutional Neural Networks, traditionally associated with image and speech processing, become a pivotal focus in MRS. Researchers explore the application of CNNs to analyse intricate patterns within music data, enhancing feature extraction and representation learning for improved recommendation accuracy. Additionally, CF transitions from a pivotal to a fundamental theme, emphasising its enduring significance in MRS research.

In the early stages of MRS, the dominant theme revolves around addressing cold start problems and incorporating context awareness. The cold start problem pertains to making accurate recommendations for new users or items with limited interaction history. Strategies to tackle this involve leveraging auxiliary data sources and contextual information, ensuring more precise recommendations for users with sparse or no historical preferences.

Simultaneously, a niche theme gaining prominence during this period is the exploration of personalisation techniques. Researchers are delving into innovative approaches to tailor recommendations based on user preferences and behaviour. This involves the development of advanced algorithms that can adapt to users' unique tastes and interests, thereby enhancing MRS's overall personalisation capabilities [92].

In the emerging themes category, personalisation takes centre stage. The focus shifts towards refining and advancing personalisation techniques, exploring novel algorithms and methodologies to provide users with highly tailored and relevant recommendations. This phase witnesses a growing interest in understanding user context, behaviour, and preferences to create more nuanced and accurate recommendation systems [93].

Conversely, Convolutional Neural Networks emerge as a pivotal focus within the basic and transversal themes. Although traditionally associated with image and speech processing, integrating CNNs in MRS becomes a fundamental theme. Researchers are exploring the application of CNNs to analyse and interpret intricate patterns within music data, leading to enhanced feature extraction and representation learning for improved recommendation accuracy [53].

In the motor themes quadrant, the spotlight is on incorporating personal characteristics into the recommendation process. This involves a deeper analysis of user attributes, demographics, and psychographics to tailor recommendations based on individual traits. Integrating personal characteristics adds a layer of sophistication to MRS, allowing for a more refined and user-centric recommendation experience [94].

In Figure 6 and Table 7, This strategic map offers a comprehensive overview of the dynamic evolution and trends in MRS research, highlighting the innovative themes that have significantly shaped the landscape over distinct periods.

4) ANALYSIS OF THE STRATEGIC DIAGRAM OF THE FORTH SUB PERIOD(2021-2023)

a: MOTOR THEMES: EMOTION RECOGNITION, AFFECTIVE COMPUTING

In the current timeframe, the strategic map significantly emphasises emotion recognition as a motor theme within MRS. Emotion recognition involves discerning an individual's emotional state, and within MRS, it introduces an innovative approach to recommending music by observing typing and clicking behaviour to interpret emotions. This method aims for higher accuracy than conventional approaches, eliminating the need for users to express their feelings explicitly. Affective computing, closely related to emotion recognition, plays a crucial role in understanding and responding to human emotions, enhancing the personalised nature of music recommendations.

b: HIGHLY DEVELOPED AND ISOLATED THEMES: ARTIFICIAL INTELLIGENCE, COLD START, EMOTION CLASSIFICATION, GROUP MUSIC RECOMMENDER, MUSIC THERAPY, NATURAL LANGUAGE PROCESSING, PERSONALIZATION, TRANSFER LEARNING

Highly developed and isolated themes in MRS encompass a broad spectrum. Artificial intelligence is a foundational theme, contributing to the overall advancement of recommendation systems. Addressing the cold-start problem, emotion classification, group music recommender systems, music therapy applications, natural language processing, personalisation, and transfer learning showcase the field's depth and diversity of focused research. Each theme has matured, signifying its pivotal role in the MRS landscape.

TABLE 6. Performance measures for the themes of the second sub period of 2015-2018.

Cluster	Callon Centrality	Callon Density	Rank Centrality	Rank Density	Cluster Frequency
collaborative filtering	0.0379	2.8571	4	1	35
music information retrieval	0.0792	7.5844	5	2	50
automatic playlist continuation	0	17.1428	2	3	12
hybrid recommendation	0	20	2	4.5	5
sentiment analysis	0	20	2	4.5	5

TABLE 7. Performance measures for the themes of the third sub period of 2019-2020.

Cluster	Callon Centrality	Callon Density	Rank Centrality	Rank Density	Cluster Frequency
collaborative filtering	0.1357	55.3846	10	7	23
music information retrieval	0.3131	8.3333	13	1	12
convolutional neural networks	0.2121	34.0909	11	3	13
personal characteristics	0.25	96.6667	12	13	14
personalisation	0	25	5	2	4
cold-start problem	0	61.6667	5	9	12
context-awareness	0	73.3333	5	12	11
lyrics	0	55.5556	5	8	7
classical music	0	62.5	5	10	4
music classification	0	50	5	5	2
music preferences	0	66.6667	5	11	6
playlist generation	0	50	5	5	2

c: EMERGING OR DECLINING THEMES: RECOMMENDATION ALGORITHM COLLABORATIVE FILTERING

The strategic map indicates a potential shift in the status of CF as a recommendation algorithm. While it has been a pivotal theme, there are signs of it experiencing changes, highlighting the dynamic nature of trends within MRS research. The decline in the number of new articles in this domain suggests a potential evolution of CF or a shift in focus toward other emerging methodologies.

d: BASIC AND TRANSVERSAL THEMES: MUSIC INFORMATION RETRIEVAL, CLUSTER-BASED, USER EXPERIENCE, CONVOLUTIONAL NEURAL NETWORKS, USER MODELING, RECOMMENDATION ALGORITHM COLLABORATIVE FILTERING

Basic and transversal themes play a fundamental role in the MRS landscape. Music Information Retrieval, cluster-based approaches, user experience considerations, CNNs, user modelling, and the continued presence of CF as a fundamental theme contribute to the foundational aspects of MRS research. These themes collectively shape music recommender systems' overall structure and functionality, providing essential elements for practical and personalised recommendations.

The strategic map, illustrated in Figure 7 and Table 8, not only highlights the transformative role of emotion recognition but also provides valuable insights into the evolution of themes within the dynamic landscape of Music Recommender Systems. The decline in new articles in certain domains reflects the maturation and consolidation of knowledge in those areas, contributing to a more prosperous and established foundation for future research.

In Figure 7 and Table 8, The strategic map highlights the pivotal placement of emotion recognition as a motor theme within the current timeframe. Emotion recognition, in simple

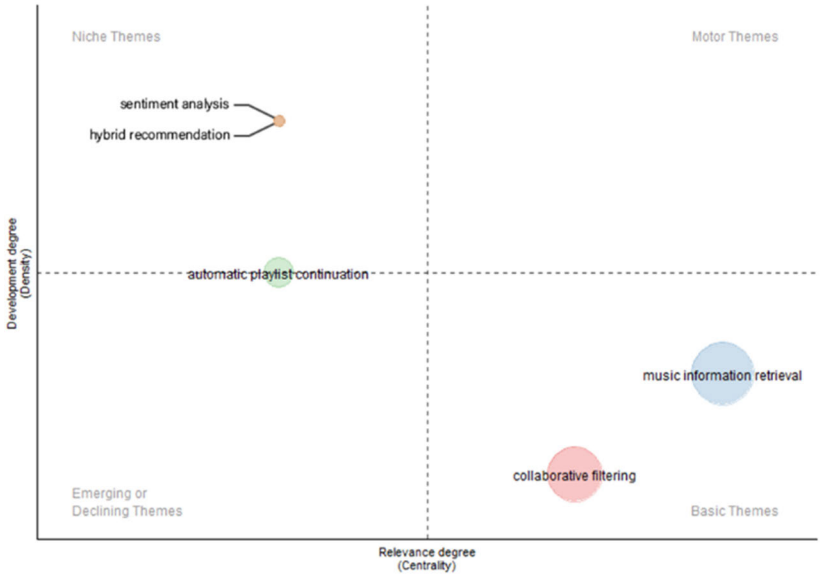


FIGURE 5. Strategic diagram for the second sub period of 2015-2018.

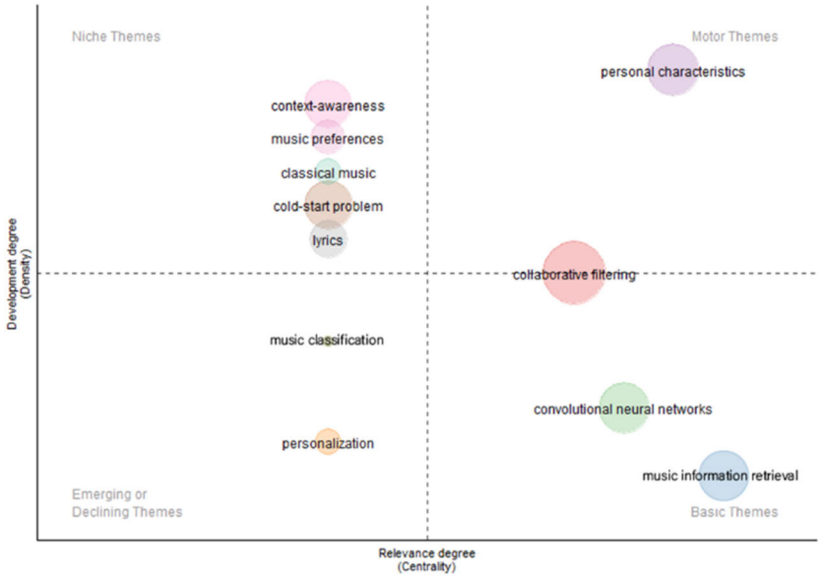


FIGURE 6. Strategic diagram for the third sub period of 2019-2020.

terms, involves discerning an individual's emotional state. Within MRS, this innovative approach recommends music by observing a person's typing and clicking behaviour, interpreting these actions to ascertain their emotions. Notably, this method aims for higher accuracy than conventional approaches, eliminating the need for users to express their feelings explicitly [95].

The diagram indicates that the decline in the number of new articles within this field may be attributed to the natural evolution of topics and keywords into foundational themes. The maturation of various aspects within MRS research has led to the consolidation of knowledge, resulting in fewer novel articles in certain domains.

This insight from the diagram underscores the transformative role of emotion recognition and the overall evolution of themes within the dynamic landscape of Music Recommender Systems.

B. THEMATIC EVOLUTION ANALYSIS

Figure 8 presents the evolution map of research within the MRS field. MIR emerges as a central thematic focus, underscoring the significance of additional information and

TABLE 8. Performance measures for the themes of the forth sub period (2021-2023).

name	N	centrality	density	rcentrality	rdensity	freq
music information retrieval	3	0.2731	25.3086	17	4	18
cluster-based	3	0.2396	28.7698	16	6	18
user experience	2	0.2222	25	15	2.5	9
convolutional neural networks	4	0.1653	26.9531	14	5	30
collaborative filtering	4	0.1417	31.7708	13	7	22
emotion recognition	3	0.1112	39.8148	11.5	17	10
affective computing	1	0.1112	33.3333	11.5	11.5	3
user modelling	1	0.0833	25	10	2.5	4
recommendation algorithm	1	0.0667	20	9	1	5
artificial intelligence	1	0	33.3333	4.5	11.5	3
cold start	1	0	33.33333	4.5	11.5	3
emotion classification	1	0	33.3333	4.5	11.5	3
group music recommender	3	0	37.0370	4.5	16	9
music therapy	1	0	33.3333	4.5	11.5	3
natural language processing	1	0	33.3333	4.5	11.5	3
personalisation	1	0	33.3333	4.5	11.5	3
transfer learning	1	0	33.3333	4.5	11.5	3

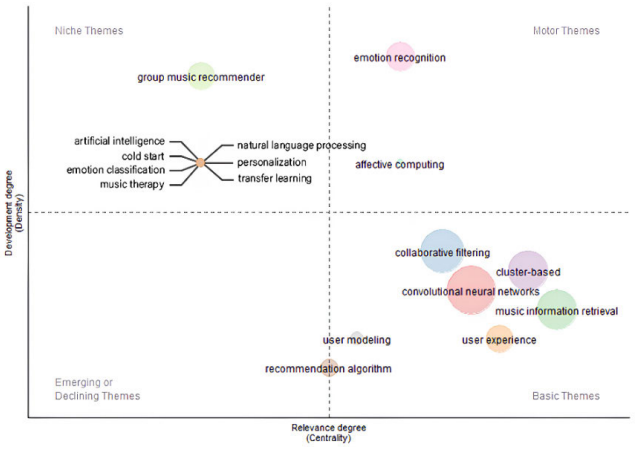


FIGURE 7. Strategic diagram for the fourth sub period of 2021-2023.

contextual insights from music and lyrics for recommender systems. Over the years, integrating Hybrid recommendation and CF has become apparent, hinting at a growing trend in hybrid approaches, including the convergence of emotion recognition and CF.

The consistent presence of CNNs in this evolution map highlights their substantial benefits in speech recognition. Their capacity to process audio data and generate latent representations without relying on traditional CF matrix factorisation positions CNNs as a remarkable tool for understanding the latent space of user behaviour. This enhances speech recognition technologies and provides valuable insights into user behaviour with contextual data from musical content [96], [97], [98], [99], [100].

As time progresses, a discernible trend toward personalisation topics becomes increasingly popular within this evolutionary map, reflecting the evolving priorities and interests within the MRS rese

Looking at how music recommendation systems have evolved, future research will likely focus on several key directions.

1. **Digitalization and Technology Integration**: The Pervasive Influence of "Digital Platforms" The relentless march of digitalisation and technology integration in music recommender systems heralds a transformative era in user experiences. As researchers delve deeper into this paradigm shift, the exploration extends to adopting cutting-edge technologies, such as convolutional neural networks, for intricate pattern recognition within music data. The continuous evolution of MIR emerges as a focal point, promising a richer understanding of musical content and context. Additionally, innovative digital tools are anticipated to reshape how users interact with and discover music.

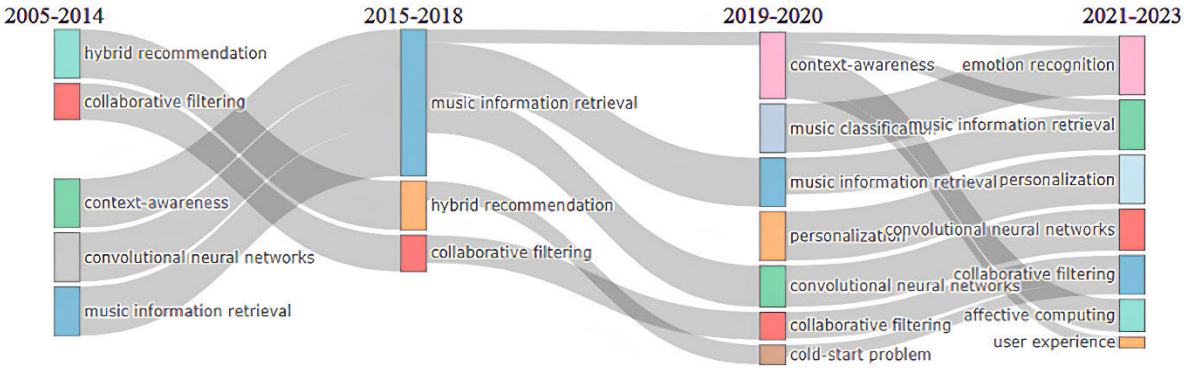


FIGURE 8. Evolution map and thematic areas for the four designated sub-periods.



FIGURE 9. Word cloud of the most used terms in music recommendation systems literature.

This forward-looking endeavour seeks to harness emerging technologies not merely for novelty but with a persistent goal – to elevate and enrich the music listening experience. The integration of these technologies is poised to go beyond conventional boundaries, offering users a more immersive, intelligent, and personalised journey through the vast landscape of musical content. The music recommender systems of the future are envisioned to seamlessly leverage the power of digital platforms, ensuring that technology becomes an enabler for a more profound, nuanced, and satisfying connection between users and their musical discoveries.

2. **Emotion Recognition**: In the realm of music recommender systems, the rising prominence of Emotion Recognition heralds a captivating frontier that transcends conventional preference-based algorithms. The future research trajectory in this domain promises a deep exploration of affective computing techniques. Emotion Recognition seeks to unravel the intricate tapestry of users' emotional states, opening doors to a more profound and personalised musical journey.
   Researchers anticipate unravelling the emotional nuances embedded within user interactions, be it typing, clicking, or other behavioural cues. This innovative approach not only aspires to understand emotions implicitly but also envisions crafting music recommendations that resonate with users on a profoundly emotional level. The ultimate goal is to surpass the limitations of traditional algorithms, offering recommendations that are not just musically apt but also aligned with the user's current emotional landscape.
   As this theme gains momentum, the integration of emotional recognition is anticipated to redefine the very essence of personalised music recommendations. The forthcoming research envisions a landscape where music recommender systems become intuitive companions capable of interpreting and responding to users' emotions, fostering a more meaningful and empathetic connection between individuals and the music they cherish.
3. **Personalization:** The enduring theme of Personalization within the realm of music recommender systems signals an unwavering commitment to tailoring the user experience with a precision that transcends generic recommendations. Future research endeavours in this area are poised to navigate the intricate landscape of advanced personalisation algorithms, delving into the

intersection of CF methods and the subtle nuances of user behaviour and musical taste.

Anticipating a paradigm shift, researchers aim to refine and enhance the accuracy of recommendations, ensuring that users encounter music that resonates intimately with their preferences. The evolution of personalisation extends beyond the mere alignment of musical genres; it involves a profound understanding of the user's context, situational preferences, and evolving tastes over time.

In the coming years, personalised music recommendations are expected to become more finely tuned, drawing from explicit user preferences and implicit signals and contextual cues. The goal is to provide users with a tailor-made music discovery experience, cultivating a sense of satisfaction and resonance that transcends the boundaries of traditional recommendation systems. As the research in Personalization advances, the vision is to offer users a journey through music that feels uniquely their own, reflective of their individuality and evolving musical identity.

4. **User Experience:** The explicit emphasis on User Experience (UX) within the evolving landscape of music recommender systems reflects an unwavering commitment to refining the interaction dynamics between users and their platforms. Future research in this realm is poised to revolve around the meticulous design of user-friendly interfaces, capturing and comprehending user feedback, and implementing novel approaches to enhance overall satisfaction and engagement.

   Anticipating a user-centric paradigm, researchers aspire to create seamless and enjoyable experiences that resonate with users on both functional and emotional levels. The design of interfaces will prioritise intuitive navigation, presenting users with an interface that feels natural and responsive to their preferences. Moreover, the integration of user feedback mechanisms will play a crucial role, ensuring that users' evolving needs and expectations are not only heard but actively incorporated into system improvements.

   As researchers delve into the User Experience domain, the focus extends beyond the functionality of recommendation algorithms to consider the holistic journey of music discovery. This involves understanding how users interact with the system, the emotions elicited during the exploration, and the creation of an environment that fosters a deeper and more meaningful connection with music. The goal is to elevate User Experience to a level where music recommender systems provide recommendations and craft an immersive and satisfying journey for each user, contributing to a lasting and positive impact on their musical exploration.
5. **Collaborative Filtering** The persistent and integral role of CF in the evolution of music recommender systems underscores an enduring interest in leveraging collective insights to enhance the precision and diversity of recommendations. Future investigations into CF will likely delve into innovative techniques, considering individual preferences, social dynamics, and group interactions.

   Anticipating a continued evolution, researchers aim to refine CF methodologies in response to the dynamic nature of user behaviours and preferences. Exploring novel CF strategies involves a deeper understanding of social connections, community influences, and the intricate interplay of shared tastes among user groups.

   In the forthcoming era of music recommendation research, collaboration extends beyond traditional user-user interactions to encompass broader social contexts. The goal is to enhance the adaptability and relevance of recommendations by acknowledging the influence of social networks and community preferences. As the CF theme advances, the vision is to create recommendation systems that reflect individual tastes and capture the richness of shared musical experiences, fostering a sense of community and shared discovery within the realm of music exploration.
6. **Social impact** Music recommender systems have the potential to impact society in diverse ways significantly. These systems, designed to recommend personalised music based on individual preferences, enrich the user experience within streaming services [101]. However, concerns regarding biases and limitations have emerged. Studies indicate that these systems often favour already popular music, potentially reinforcing the dominance of well-known artists and limiting exposure to lesser-known musicians [102]. The societal impact extends beyond the music industry, as exemplified by YouTube's video recommendation system, which is criticised for its role in fostering echo chambers, filter bubbles, polarisation, and the spread of disinformation [103]. Ethical challenges must be addressed to ensure fair and beneficial outcomes for users and society, including transparency and explainability [104].

Despite these concerns, music recommender systems hold the potential to bridge the gap between popular and niche music preferences. Innovations like gene feature fusion, CF in Spark architecture, and deep learning algorithms address the traditional challenge of neglecting non-mainstream music enthusiasts. These advancements enhance the accuracy and diversity of music recommendations, contributing to a more vibrant musical landscape where artists with diverse styles and genres gain recognition, benefiting listeners and creators.

Moreover, the societal applications of music recommender systems extend beyond individual users to influence collaborative and social aspects of music consumption. Incorporating contextual information, such as location, emotion, and user activities, paves the way for developing

context-aware music recommender systems. This evolution in recommendation strategies holds promise in shaping shared musical experiences, allowing users to discover and enjoy music in specific contexts or moods. The societal impact encompasses fostering communal music discovery, shared playlists, and collective engagement with music content.

In conclusion, the societal impact and applications of Music Recommendation Systems are multifaceted. They enhance individual music exploration and contribute to a more inclusive and diverse music culture. The evolving approaches in recommendation strategies and the integration of contextual information demonstrate the potential for these systems to shape not just individual preferences but also collective music experiences, influencing societal trends and fostering a richer musical ecosystem.

VI. FUTURE WORKS

A. FUTURE DIRECTIONS FOR RESEARCH

1. Exploring Theoretical Advancements: Probe the ongoing theoretical landscape to uncover gaps in understanding user experiences within music recommender systems. Evaluate the potential integration of emerging theoretical frameworks, such as deep learning paradigms, in enriching the comprehension of intricate musical patterns. Examine the evolution of theoretical constructs in music information retrieval to gain a more nuanced understanding of musical content and context.
2. Advancing Methodological Approaches in Emotion Recognition: Delve into methodological innovations within affective computing to decode the intricate tapestry of users' emotional states. Explore novel methodological approaches for implicitly deciphering users' emotions, incorporating behavioural cues like typing and clicking. Develop methodologically robust recommendation models that factor in emotional nuances, providing a more profound and personalised musical journey.
3. Refining Methodological Techniques in Personalization: Investigate advanced methodological algorithms that transcend conventional musical genre alignment, incorporating implicit signals and contextual cues. Explore the intersection of methodological approaches, particularly CF methods, and user behaviour to enhance the precision of recommendations. Focus on adapting methodological strategies to dynamic shifts in user tastes, situational preferences, and evolving contextual dynamics.
4. Advancing User Experience Methodologies (UX): Design methodologically sound interfaces prioritising intuitive navigation and responsiveness to user preferences. Implement robust methodologies for capturing and comprehending user feedback, facilitating continuous system improvements. Explore innovative methodological approaches to elevate user experience, ensuring a seamless and enjoyable journey in music discovery.
5. Advancing Methodological Frontiers in CF: Explore innovative methodological approaches in CF that consider individual preferences, social dynamics, and group interactions. Investigate the impact of social networks and community preferences on methodological frameworks in CF to enhance adaptability and relevance. Extend methodological collaboration beyond traditional user-user interactions to encompass broader social contexts, nurturing a sense of community in music exploration.

B. EMERGING IDEAS AND CONCEPTIONS BEYOND CURRENT TRENDS

In addition to the outlined future directions, it's imperative to acknowledge the potential for paradigm-shifting innovations that transcend current trends. Emerging ideas and conceptions could include:

1. **Integrating Neuroscientific Principles:** We are exploring integrating neuroscientific principles into music recommender systems to decode brain responses to music and enhance personalisation based on cognitive and emotional states.
2. **Blockchain Technology:** Leveraging blockchain technology to create decentralised music recommendation platforms, empowering artists and ensuring fair compensation while fostering community-driven curation.
3. **Multi-modal Recommendation Systems:** Develop recommendation systems that leverage multiple modalities such as audio, lyrics, and music videos to provide a holistic and immersive music discovery experience.
4. **Ethical and Fairness Frameworks:** Prioritizing the development of ethical frameworks and fairness metrics to address biases, promote diversity, and ensure transparent and accountable music recommendations.

These emerging ideas hold the potential to reshape the landscape of music recommender systems, ushering in new possibilities for personalised, inclusive, and ethically-driven music exploration experiences.

C. CONCLUSION AND FUTURE METHODOLOGICAL OUTLOOK

In conclusion, the evolution of Music Recommender Systems has witnessed dynamic shifts and emerging trends. As the field progresses, future research should focus on exploring theoretical advancements, refining methodological approaches in emotion recognition and personalisation, advancing methodologies for user experience, and innovating CF strategies. Researchers in this domain are encouraged to engage in methodologically rigorous investigations to contribute to ongoing advancements in Music Recommender Systems research, ensuring a robust theoretical and methodological foundation for the evolving needs of users.

VII. CONCLUSION

In the dynamic realm of MRS, a meticulous bibliometric analysis provides insights into an ever-evolving landscape that mirrors broader market dynamics. The exploration of pivotal themes—ranging from CF and music information retrieval to CNN-based recommendations, context-aware systems, and personalisation—unveils the dynamic shifts and emerging concepts shaping the trajectory of Music Recommender Systems research. Spanning the years from 2005 to 2023, an examination of 791 publications forms the foundation for this comprehensive investigation into the evolutionary trends within the Music Recommender Systems domain.

The analysis discerns a recent dip in the overall interest in Music Recommender Systems research over the past three years. Despite this decline, attention has markedly gravitated towards context-aware models and the integration of deep learning methodologies. Notably, CF maintains its prominence as a critical focus, driven by the expansive user base of music streaming services and the wealth of data from user interactions, reinforcing its pivotal role in the recommender systems landscape.

Furthermore, an intriguing revelation surfaces in the underdeveloped theme surrounding hybrid approaches for contextually aware systems. This theme, characterised by its demand for lower computational resources, signifies a promising avenue for future research. Creating more personalised systems tailored to individual users through innovative hybrid models represents an exciting frontier that researchers could explore in the evolution of Music Recommender Systems.

Music recommender systems have the potential to impact society significantly in diverse ways. These systems, designed to recommend personalised music based on individual preferences, enrich the user experience within streaming services [101]. However, concerns regarding biases and limitations have emerged. Studies indicate that these systems often favour already popular music, potentially reinforcing the dominance of well-known artists and limiting exposure to lesser-known musicians [102]. The societal impact extends beyond the music industry, as exemplified by YouTube's video recommendation system, which is criticised for its role in fostering echo chambers, filter bubbles, polarisation, and the spread of disinformation [103]. Ethical challenges must be addressed to ensure fair and beneficial outcomes for users and society, including transparency and explainability [104].

Despite these concerns, music recommender systems hold the potential to bridge the gap between popular and niche music preferences. Innovations like gene feature fusion, CF in Spark architecture, and deep learning algorithms address the traditional challenge of neglecting non-mainstream music enthusiasts. These advancements enhance the accuracy and diversity of music recommendations, contributing to a more vibrant musical landscape where artists with diverse styles and genres gain recognition, benefiting listeners and creators.

Moreover, the societal applications of music recommender systems extend beyond individual users to influence collaborative and social aspects of music consumption. Incorporating contextual information, such as location, emotion, and user activities, paves the way for developing context-aware music recommender systems. This evolution in recommendation strategies holds promise in shaping shared musical experiences, allowing users to discover and enjoy music in specific contexts or moods. The societal impact encompasses fostering communal music discovery, shared playlists, and collective engagement with music content.

In conclusion, the societal impact and applications of Music Recommendation Systems are multifaceted. They enhance individual music exploration and contribute to a more inclusive and diverse music culture. The evolving approaches in recommendation strategies and the integration of contextual information demonstrate the potential for these systems to shape not just individual preferences but also collective music experiences, influencing societal trends and fostering a richer musical ecosystem.

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